

Tutorial III

Artificial Intelligence (CST-5105)

I. Choose the correct answer. (5 marks)

1. What is the solution of the prolog query, 'larger(9,5,X).' with the two given clauses, 'larger(A,B,A):-A>B.' and 'larger(A,B,B).'?
 (a) X=9 (b) X=9; X=5 (c) true (d) false
2. Uncertainties in logic are improved by usinglogic.
 (a) first-order (b) fuzzy (c) propositional (d) predicate
3. A list is written in Prolog usingto enclose the items.
 (a) curly braces (b) single quotes
 (c) square brackets (d) double quotes
4. Which of the following is true for prolog query?
 (a) not (2=4) (b) \+(2=4) (c) none (d) both
5. Which one from the options would return true/yes for given prolog program?
 Boy (john, 123).
 girl (jane, 234).
 student (john, 123).
 (a)?- girl (jane, x). (b)?- boy('john',123).
 (c) All of above. (d) None of above.

II. (a) Let **Leuko** be the “Leukocyte value higher than 1000.” and **App** be “Patient has appendicitis (appendix inflammation)”. By using the probabilities given in the table, calculate the following. (5 marks)

- (i) P(Leuko)
- (ii) P(Leuko|App)
- (iii) P(App|Leuko)
- (iv) P(App)

P(App, Leuko)	App	¬App	total
Leuko	0.23	0.31	0.54
¬Leuko	0.05	0.41	0.46
total	0.28	0.72	1

(b) Consider the following Bayesian network, where F = having the flu and C = coughing: (5 marks)



Write down the joint probability table specified by the Bayesian network.

Determine the probabilities for the Bayesian network.

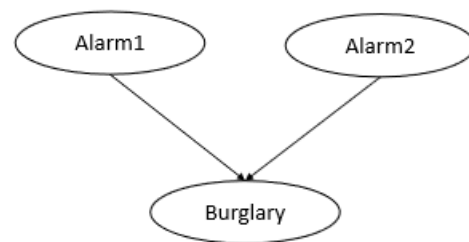
P (C), P (F|C), P (F| ¬ C)

- (c) A patient has been suffering from shortness of breath (called dyspnoea) and visits the doctor, worried that he has lung cancer. The doctor knows that other diseases, such as tuberculosis and bronchitis, are possible causes, as well as lung cancer. She also knows that other relevant information includes whether or not the patient is a smoker (increasing the chances of cancer and bronchitis) and what sort of air pollution he has been exposed to. Construct a Bayesian Network for this scenario and develop the corresponding mode. **(5 marks)**

- III. (a) To safeguard your house, you recently installed alarm system by reputable manufacturers that use completely different sensors for their alarm system. Alarm1 means that the first alarm system rings, Alarm2 means that the second alarm system rings, and Burglary means that a burglary is in progress. Consider the following Bayesian network.

Assume that:

$$\begin{aligned}P(\text{Alarm1}) &= 0.1 \\P(\text{Alarm2}) &= 0.2 \\P(\text{Burglary} \mid \text{Alarm1}, \text{Alarm2}) &= 0.8 \\P(\text{Burglary} \mid \text{Alarm1}, \neg \text{Alarm2}) &= 0.7 \\P(\text{Burglary} \mid \neg \text{Alarm1}, \text{Alarm2}) &= 0.6 \\P(\text{Burglary} \mid \neg \text{Alarm1}, \neg \text{Alarm2}) &= 0.5\end{aligned}$$



Calculate the following conditional probabilities $P(\text{Alarm2} \mid \text{Burglary}, \text{Alarm1})$. Show all of your reasoning. **(5marks)**

- (b) For given English statements write a prolog program. **(5marks)**

- Facts & Rules

- (i) jia is a woman.
- (ii) john is a man.
- (iii) john is healthy.
- (iv) jia is healthy.
- (v) john is wealthy.
- (vi) anyone is a traveler if he is healthy and wealthy.
- (vii) anyone can travel if he is a traveler.

- Goals.

- (1) Who can travel?
- (2) Who is healthy and wealthy?

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